Kalasalingam Acadmey of Research and Education

StudyBuddy: AI-Powered Study Assistant Using Retrieval-Augmented Generation (RAG)

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Project Trainer / Guide

Mr. SHIRWIN PRINCE I

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1. Introduction

Artificial Intelligence (AI) is transforming the education sector by providing intelligent tools that support personalized learning, automated assessment and interactive knowledge assistance. Students increasingly depend on digital resources such as PDF documents, lecture notes, textbooks and online study materials for academic preparation.

However, finding specific information from lengthy academic documents can be time-consuming. Traditional search methods mainly depend on matching keywords and may not understand the actual meaning of a student's question.

To address these challenges, StudyBuddy is developed as an AI-powered intelligent learning assistant that combines Retrieval-Augmented Generation (RAG) with Large Language Models (LLMs).

The application allows administrators to upload academic materials and organize them according to subjects. Students can interact with these materials by asking questions in natural language and receiving AI-generated answers based on relevant document content.

In addition to question answering, the application provides AI-generated multiple-choice quizzes, quiz evaluation, chat history and basic learning progress information.

The primary purpose of StudyBuddy is to provide a centralized and interactive learning environment that helps students access academic information and improve their preparation.

2. Problem Statement & Objectives

Problem: Students frequently depend on lengthy PDF documents, lecture notes and other academic materials for examination preparation. Finding specific information from these resources can be difficult and time-consuming.

Traditional document search systems mainly identify matching keywords and may not understand the meaning or context of a student's question.

Additionally, students may need to prepare practice questions manually and maintain separate records of their learning activities.

General-purpose AI chatbots can provide explanations, but their responses may not always be aligned with the specific academic materials provided by an institution or instructor.

Therefore, there is a need for an intelligent learning system that can understand uploaded study materials, retrieve relevant information, generate context-based answers and support student assessment.

StudyBuddy addresses these challenges through a Retrieval-Augmented Generation architecture combined with AI-powered quiz generation and learning history management.

Objectives:

- To develop an intelligent AI-based learning assistant.
- To provide document-based question answering.
- To implement semantic search using vector embeddings.
- To integrate ChromaDB for document retrieval.
- To integrate Google Gemini for answer generation.
- To provide automatic MCQ quiz generation.
- To maintain student chat and quiz history.
- To simplify academic information retrieval.

3. System Architecture (RAG Pipeline)

The architecture of StudyBuddy is divided into two primary phases: Data Ingestion and Query Processing.

3.1 Data Ingestion Phase

The Data Ingestion Phase is responsible for processing academic documents uploaded by administrators.

1. Document Upload: Administrators upload academic documents in supported formats such as PDF, DOCX and TXT.

2. Text Extraction: The system extracts textual information from the uploaded documents.

3. Text Chunking: The extracted text is divided into smaller meaningful segments using a custom word-based chunking method.

4. Embedding Generation: Sentence Transformers using the all-MiniLM-L6-v2 model converts text chunks into numerical vector representations.

5. Vector Storage: The generated embeddings and associated metadata are stored in ChromaDB for efficient semantic retrieval.

3.2 Query & Generation Phase

When a student submits a question, the system processes the query using the same embedding model.

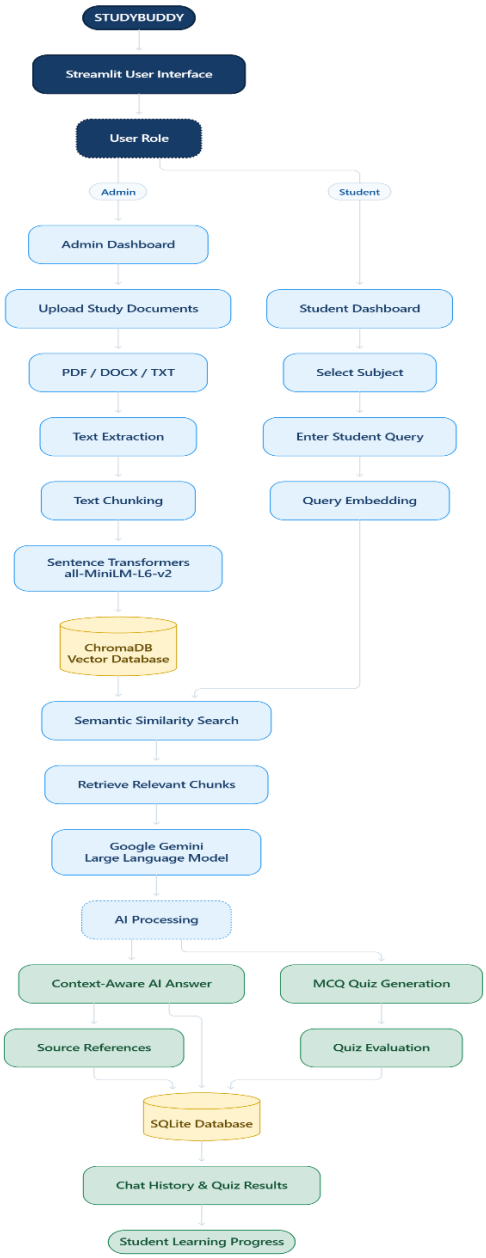
The query embedding is compared with stored document embeddings in ChromaDB through semantic similarity search. The system retrieves the five most relevant document chunks by default.

The retrieved information is provided as contextual input to Google Gemini, which generates a relevant response based on the available study material.

The generated answer and available source references are displayed to the student through the Streamlit interface.

The same retrieval architecture also supports AI-powered multiple-choice quiz generation using relevant academic content.

3.4 RAG- Architecture



4. Implementation Details & Tech Stack

StudyBuddy is implemented using Python and Streamlit, with additional AI and database technologies for document processing, semantic retrieval and response generation.

The following technologies are used in the application:

Component	Technology Used	Purpose
Programming Language	Python	Core application logic and Backend processing
Frontend Framework	Streamlit	Developing the interactive web interface
AI Orchestration	Custom Python RAG Pipeline	Managing document processing, retrieval and generation
Vector Database	ChromaDB	Storing and retrieving document embeddings
Embedding Model	Sentence Transformers (all-MiniLM-L6-v2)	Converting text into numerical vector representations
Large Language Model	Google Gemini	Generating context-aware answers and MCQs
Relational Database	SQLite	Managing users, subjects, chat history and quiz results
Document Processing	Document Processing	Extracting text from academic documents
Deployment	Streamlit Community Cloud	Hosting the application online
Version Control	Git and GitHub	Source-code management and collabration

4.1 Sample Chunking Configuration

To improve document retrieval, the extracted text is divided into smaller chunks using a custom word-based chunking method.
The implementation uses the following configuration:

- Chunk Size: 900 words.
- Chunk Overlap: 150 words.

The overlapping content helps preserve contextual continuity between consecutive text segments.

4.2 Database Management

StudyBuddy uses two database technologies for different purposes.
SQLite: Stores structured application information such as user accounts, subjects, uploaded document records, chat history and quiz results.
ChromaDB: Stores document embeddings and associated metadata to support semantic similarity search and relevant content retrieval.

5. Results and Testing

The StudyBuddy application provides an interactive environment for students to access academic study materials and interact with an AI-powered learning assistant.

The application supports administrator and student authentication, subject management, document uploading, document processing, AI-based question answering, source references, quiz generation and evaluation, chat history and student progress tracking.

Functional testing is performed to verify the major application workflows.

The following test scenarios are considered:

Test Case	Description	Expected Result
TC01	Admin Authentication	Successful administrator login
TC02	Student Registration	Student account creation
TC03	Subject Management	Successful subject processing
TC04	Document Upload	Successful document processing
TC05	AI Question Answering	Content-aware answer processing
TC06	Soucre References	Display of document references
TC07	Quiz Generation	Successful MCQ generation

TC08	Quiz Evaluation	Correct score calculation
TC09	Chat History	Retrieval of previous conversations
TC10	Progress Dashboard	Display of student learning information

The testing results are recorded based on the actual execution of each test case.

The application demonstrates the practical integration of Retrieval-Augmented Generation and generative AI for academic learning assistance.

6. Conclusion & Future Scope

Conclusion: StudyBuddy is an AI-powered intelligent learning assistant designed to simplify academic learning through document-based question answering and automated assessment.

The application combines Retrieval-Augmented Generation, Sentence Transformers, ChromaDB and Google Gemini to retrieve relevant academic information and generate context-aware responses.

The system provides separate administrator and student interfaces, allowing academic document management, interactive question answering, quiz generation, evaluation and learning history management.

By integrating semantic search with generative AI, StudyBuddy provides an interactive approach to accessing and understanding academic study materials.

Future Scope:
Future improvements to StudyBuddy may include:

- OCR support for extracting text from scanned PDF documents.
- Multilingual question answering and learning assistance.
- AI-powered flashcard generation.
- Personalized study timetable generation.
- Advanced student performance analytics.
- Improved document retrieval and ranking techniques.
- Additional interactive learning and assessment features.